INCH-POUND

MIL-T-82846(OS)
5 February 1990
SUPERSEDING
WS 20886
8 November 1983

MILITARY SPECIFICATION

THREAD, SPLICING, NYLON

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of twisted soft multiple cord nylon thread for machine- and hand-sewing, referred to herein as the thread.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-T-295 Thread, Nylon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material requirements.** The thread shall conform to V-T-295, Type I, Class A.

3.3 **Maximum percent extractable material.** The maximum allowable extractable material shall be 3.5 percent.

3.4 **Infrared (IR) identification of extractable material.** The extractable material infrared spectrum shall conform qualitatively to the identification shown on figure 1.

3.5 **Workmanship.** Workmanship shall be such that the thread is of uniform high quality and free from defects that would render it unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

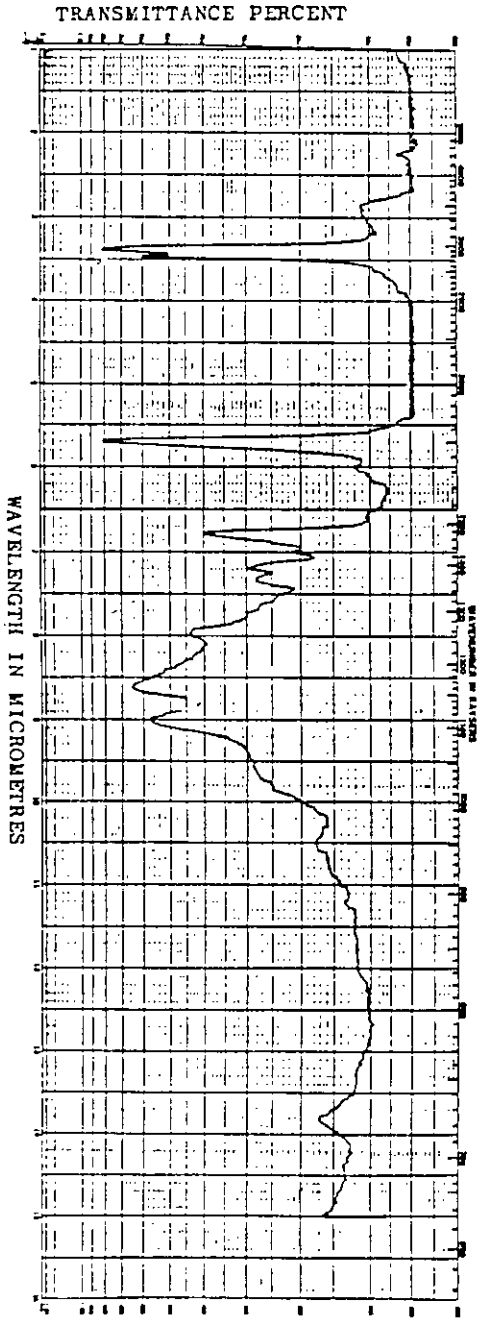


FIGURE 1. Infrared spectrum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4) and such other additional tests as may be required to prove that the thread is suitable for its intended use.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests and examinations specified in table I.

TABLE I. Quality conformance inspection.

Tests and Examinations	Requirement	Test Method
Visual examination	3.6 and Section 5	4.5.1
Percent extractable material	3.3	4.5.2
Infrared identification	3.4	4.5.3

4.4.1 **Sampling.** The number of spools of thread (samples) to be selected shall be in accordance with table II.

TABLE II. Sampling.

Spools in Lot	Number of Samples to be Selected for Testing
1 - 3	All
4 - 8	2
9 - 15	3
16 - 25	5
26 - 50	8
51 - 90	13
91 - 150	20

4.4.2 **Lot.** A lot shall consist of a maximum of 150 spools of less than 2 pounds each and shall be from one manufacturing run.

4.4.2.1 **Selection of material.** Material shall be taken from any place within the spool in sufficient quantity to perform all tests specified herein.

4.4.2.2 **Test specimens.** For the extractable material test, a minimum of three specimens shall be tested from each sample. The average results of a minimum of three test specimens shall be reported in addition to reporting the individual test specimen value. The average results shall conform to 3.2. For IR identification, a minimum of two samples shall be randomly selected from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

samples selected as specified in table II. A minimum of one IR identification shall be performed for each sample.

4.5 **Test methods.** The following test methods and procedures shall be used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample shall be visually examined for conformance to 3.5. Each container shall be examined for conformance to section 5. Contractor certification verifying conformance to 3.2 shall be required.

4.5.2 **Percent extractable material.** The percent extractable material shall be determined in accordance with the following:

a. Place 3 $\pm$ 1 grams (g) of thread sample, weighed to the nearest 0.001 g (W1), into a fat-free cellulose, alundum, or fritted glass soxhlet extraction thimble previously extracted with solvent, and place in a soxhlet extraction apparatus.

b. Extract the thread with acetone for a minimum of 4 hours. Adjust the heat so the siphoning takes place three to five times per hour.

c. Transfer the extract and solvent from the extraction flask into a previously tared beaker (W2), rinsing with a small quantity of solvent to ensure complete transfer.

d. Evaporate the solvent from the extract in a drying oven at 50 $\pm$ 3°C. Dry to a constant weight and weigh the beaker and residue to the nearest 0.001 g (W3).

e. Calculate the percent extractable material as follows:

$$\text{Percent extractable material} = \frac{(W3 - W2)}{W1} \times 100$$

Where: W1 = Thread sample weight, g

W2 = Tared beaker weight, g

W3 = Weight of beaker and dried residue, g

4.5.3 **IR identification of extractable material.** The IR identification of extractable material shall be determined in accordance with the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

a. Obtain extract from percent extractable materials test (see 4.4.2).

b. Dissolve in spectrophotometric acetone, and evaporate on a sodium chloride disc.

c. Scan from 2 to 15 micrometers on a double beam spectrophotometer.

d. Compare the scan qualitatively with figure 1 and report as "accept" or "reject."

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The thread shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Lot number, material name, and number of spools.

d. Supplier name and lot number on inside diameter of each spool.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The thread is intended for use in stitching preimpregnated cloth phenolic materials.

6.2 Acquisition requirements. Acquisition documents must specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 2.1.2, and 2.2).

c. Whether first article inspection is required (see 3.1).

d. Special marking, if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N813)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82846	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
THREAD, SPLICING, NYLON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER NAVAL ORDNANCE STATION	b. TELEPHONE *(Include Area Code)* (1) Commercial: (301)743-4510 (2) AUTOVON: 364-4510
c. ADDRESS *(Include Zip Code)* ATTN: CODE 3710 INDIAN HEAD, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil - Downloaded: 2026-10-23T07:29Z
Check the source to verify that this is the current version before use.